UNIVERSAL INJECTIVITY OF A MODULE MAP IS EQUIVALENT TO THE FUNCTOR C GIVING A SPLIT SURJECTION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs, for the adjunction identity and for the faithfulness lemma; both are named rather than reproduced and the author's use of each is written out.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- The adjunction identity relating Hom into C and the functor C applied to a tensor product. Same chapter, equation-adjunction, invoked at proof line 1225. Supplies the commutative square with isomorphic vertical arrows.
- Faithfulness of the functor C. Same chapter, Lemma lemma-C-is-faithful, invoked at proof line 1242. Supplies the converse direction from surjectivity of the rows to injectivity of the base change.
- Universal injectivity of a module map. Same chapter, the defining property used in the statement at line 1220. Supplies the property being characterized.

Source reference. Original equation 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let R be a ring. A morphism $f : M \to N$ in Mod_R is universally injective if and only if $C(f) : C(N) \to C(M)$ is a split surjection.*

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Proof. By (0), for any $P \in \mathrm{Mod}_R$ we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}_R(P, C(N)) & \xrightarrow[\mathrm{Hom}_R(P, C(f))]{} & \mathrm{Hom}_R(P, C(M)) \\ \downarrow \cong & & \downarrow \cong \\ C(P \otimes_R N) & \xrightarrow{C(1_P \otimes f)} & C(P \otimes_R M). \end{array}$$

If f is universally injective, then $1_{C(M)} \otimes f : C(M) \otimes_R M \to C(M) \otimes_R N$ is injective, so both rows in the above diagram are surjective for $P = C(M)$. We may thus lift $1_{C(M)} \in \mathrm{Hom}_R(C(M), C(M))$ to some $g \in \mathrm{Hom}_R(C(N), C(M))$ splitting $C(f)$. Conversely, if $C(f)$ is a split surjection, then both rows in the above diagram are surjective, so by Lemma 0, $1_P \otimes f$ is injective. $\square$